# Solve with us 6.2 - Not Graded

This assignment will not be graded and is only for practice.
Key Points: 1Key Points: 1

- Let $T: V \to W$ be a linear transformation. We define kernel of $T$ (denoted by $ker(T)$) and image of $T$ (denoted by $Im(T)$) as follows:

$$Ker(T) = \{v \in V \mid T(v) = 0\}$$
$$Im(T) = \{w \in W \mid \text{ there exists } v \in V \text{ for which } T(v) = w\}$$

- $T$ is injective (i.e., $T$ is a monomorphism) if and only if $Ker(T) = \{0\}$.
- A linear transformation $T: V \to W$ is surjective (i.e., $T$ is epimorphism) if and only if $Im(T) = W$.
- $Nullity(T) = dim(Ker(T))$ and $Rank(T) = dim(Im(T))$.
- Let $T: V \to W$ be a linear transformation. Then

$$Rank(T) + Nullity(T) = dim(V).$$

  This is known as the rank nullity theorem.

- Let $T: V \to W$ be a linear transformation. $T$ is injective if and only if $Nullity(T) = 0$. $T$ is surjective if and only if $Rank(T) = dim(W)$.

Consider the following linear transformation:

$$T: \mathbb{R}^3 \to \mathbb{R}^2$$
$$T(x, y, z) = (2x + 3z, 4y + z)$$

Answer the following questions:

*1 point*

Which of the following Options is true?

[Hint: $T(x, y, z) = (2x + 3z, 4y + z) = (0, 0)$, i.e., $2x + 3z = 0$ and $4y + z = 0$]

○ $ker(T) = \{(3y, y, -4y) \mid y \in R\}$

○ $ker(T) = \{(6y, y, 4y) \mid y \in R\}$

○ $ker(T) = \{(6y, y, -4y) \mid y \in R\}$

○ $ker(T) = \{(3y, y, 4y) \mid y \in R\}$

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$ker(T) = \{(6y, y, -4y) \mid y \in R\}$

*1 point*

Which of the following is a basis of $Ker(T)$?

○

$\{(3,1,-4)\}$

○ $\{(6,0,-4),(0,1,-4)\}$

○ $\{(6,1,-4)\}$

○ $\{(3,0,-2),(0,1,-4)\}$

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$\{(6,1,-4)\}$

Find the nullity of $T$.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) 1

*1 point*

Find the rank of $T$.

[Hint: use rank nullity theorem.]

**No, the answer is incorrect.**
**Score: 0**
**Feedback:**
– Here dim(V) = 3 as V = $\mathbb{R}^3$.
**Accepted Answers:**
(Type: Numeric) 2

*1 point*

*1 point*

Which of the following is true?

○ $T$ is both injective and surjective.

○ $T$ is injective but not surjective.

○ $T$ is surjective but not injective.

○ $T$ is neither injective nor surjective.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$T$ is surjective but not injective.

*1 point*

Which option represents the kernel and image of the following linear transformation?

$$T: R^2 \to R^2$$
$$T(x,y) = (x,0)$$

○ $ker(T) = Span\{(1,0)\}, Im(T) = Span\{(1,0)\}$.

- ○ $ker(T) = Span\{(1,0)\}, Im(T) = Span\{(0,1)\}$.
- ○ $ker(T) = Span\{(0,1)\}, Im(T) = Span\{(1,0)\}$.
- ○ $ker(T) = Span\{(0,1)\}, Im(T) = Span\{(0,1)\}$.

**No, the answer is incorrect.**
**Score: 0**
**Feedback:**

• Ker(T) = {(0, y) | y ∈ R}.

**Accepted Answers:**

$ker(T) = Span\{(0,1)\}, Im(T) = Span\{(1,0)\}$.

Key Points: 2

- Let $T: V \to W$ be a linear. Follow the steps below to find bases for the null space and the range space of $T$.

Step 1: Find the matrix $A$ corresponding to $T$ with respect to some standard ordered bases $\beta = \{v_1, v_2, ..., v_n\}$ and $\gamma = \{w_1, w_2, ..., w_m\}$ for $V$ and $W$ respectively.
Step 2: Use row reduction on $A$ to obtain the matrix $R$ which is in reduced row echelon form.

Step 3: The basis of the solution space of $Rx = 0$ is the basis of null space of matrix $A$ and can be obtained by finding the pivot and non-pivot columns (dependent and independent variables) as seen earlier.

Step 4: The vectors $\begin{bmatrix} c_{11} \\ c_{12} \\ \vdots \\ c_{1n} \end{bmatrix}, \begin{bmatrix} c_{21} \\ c_{22} \\ \vdots \\ c_{2n} \end{bmatrix}, ..., \begin{bmatrix} c_{k1} \\ c_{k2} \\ \vdots \\ c_{kn} \end{bmatrix}$ form a basis of the null space of $A$ precisely when the vectors $v'_1, v'_2, ..., v'_k \in \ker(T)$ where $v'_i = \sum_{j=1}^{n} c_{ij} v_j$, form a basis for $\ker(T)$. Use the basis obtained in step 3 to thus get a basis for $\ker(T)$.

Step 5: Recall that if $i_1, i_2, \ldots, i_r$ are the columns of $R$ containing the pivot elements, then the same columns of $A$ form a basis for the column space of $A$.

Step 6: The vectors $\begin{bmatrix} d_{11} \\ d_{12} \\ \vdots \\ d_{1m} \end{bmatrix}, \begin{bmatrix} d_{21} \\ d_{22} \\ \vdots \\ d_{2m} \end{bmatrix}, ..., \begin{bmatrix} d_{r1} \\ d_{r2} \\ \vdots \\ d_{rm} \end{bmatrix}$

form a basis of the column space of $A$ precisely when the vectors $w_1', w_2', \ldots, w_r' \in im(T)$ where $w_i' = \sum_{j=1}^{m} d_{ij} w_j$, form a basis for $im(T)$. Use the basis obtained in step 5 to thus get a basis for $im(T)$.

Consider the following linear transformation:

$$T : R^3 \to R^3$$
$$T(x, y, z) = (x - z, 2x + 3y + z, 3y + 3z)$$

Answer the following questions 7-9 using the information given above.

*1 point*

Which of the matrices corresponds the given linear transformation $T$ with respect to the standard ordered basis of $R^3$ for both the domain and the codomain?

○

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 3 \\ -1 & 1 & 3 \end{bmatrix}$$

○

$$\begin{bmatrix} 1 & 2 & 3 \\ -1 & 3 & 3 \\ 0 & 1 & 0 \end{bmatrix}$$

○

$$\begin{bmatrix} 1 & -1 & 0 \\ 2 & 3 & 1 \\ 3 & 3 & 0 \end{bmatrix}$$

○

$$\begin{bmatrix} 1 & 0 & -1 \\ 2 & 3 & 1 \\ 0 & 3 & 3 \end{bmatrix}$$

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**

$$\begin{bmatrix} 1 & 0 & -1 \\ 2 & 3 & 1 \\ 0 & 3 & 3 \end{bmatrix}$$

*1 point*

What will be the kernel of $T$?

○

$\{(x, 0, z), (0, y, z) \mid x, y, z \in R\}$.

○

$\{(z, 0, z), (0, -z, z) \mid z \in R\}$.

○

$\{(z, -z, z) \mid z \in R\}$.

- $\{(x, 0, z), (0, -x, z) \mid x, z \in R\}$.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$\{(z, -z, z) \mid z \in R\}$.

*1 point*
What will be the basis of the image space of $T$ found by the algorithm mentioned in the Key Points ?

- $\{(1, 0, -1), (2, 3, 1)\}$
- $\{(1, 0, 0), (0, 1, 0)\}$
- $\{(1, 2, 0), (0, 3, 3)\}$
- $\{(-1, 1, 3)\}$

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$\{(1, 2, 0), (0, 3, 3)\}$

Check Answers
Your score is: 0/9